Probability

ANSWER KEY

Section 1: Discrete Random Variables & Variance

1 A discrete random variable X has distribution: $P(X=0)=0.1$, $P(X=1)=0.4$, $P(X=2)=0.5$. What is $E(X)$?

a. 1.4 ✓

b. 1.0

c. 1.5

d. 1.2

$E(X)=0(0.1)+1(0.4)+2(0.5)=0.4+1.0=1.4$.

2 For the above variable X, what is the variance $\text{Var}(X)$?

a. 0.44 ✓

b. 0.50

c. 0.64

d. 0.84

$E(X^2)=0^2(0.1)+1^2(0.4)+2^2(0.5)=0.4+2.0=2.4$. $\text{Var}(X)=2.4-(1.4)^2=2.4-1.96=0.44$.

3 If X is a binomial random variable with parameters $n=100$ and $p=0.2$, what is the standard deviation σ?

a. 4 ✓

b. 16

c. 20

d. 2

$\sigma=\sqrt{np(1-p)}=\sqrt{100(0.2)(0.8)}=\sqrt{16}=4$.

Section 2: Continuous Distributions & Probabilities

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4 A continuous random variable X is uniformly distributed over the interval $[2, 10]$. What is $P(4 \le X \le 7)$?

a. $\frac{3}{8}$ ✓

b. $\frac{1}{2}$

c. $\frac{5}{8}$

d. $\frac{1}{4}$

Probability density $f(x) = \frac{1}{10-2} = \frac{1}{8}$. $P(4 \le X \le 7) = \frac{7-4}{8} = \frac{3}{8}$.

5 What is the expected value of a uniform random variable on $[a, b]$?

a. $\frac{a+b}{2}$ ✓

b. $\frac{b-a}{2}$

c. $\frac{a+b}{12}$

d. ab

The center of mass of a uniform distribution is its midpoint: $\frac{a+b}{2}$.

6 In any continuous probability distribution, what is $P(X = c)$ for any single exact point c?

a. 0 ✓

b. 1

c. $\frac{1}{2}$

d. Undefined

For any continuous distribution, the probability of taking any single exact real value is 0 because the area of a single line is zero.

Section 3: Conditional Probability & Joint Events

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7 If $P(A) = 0.5$ and $P(B) = 0.4$, and A and B are mutually exclusive, what is $P(A \cap B)$?

a. 0 ✓

b. 0.20

c. 0.90

d. 0.10

Mutually exclusive events cannot occur simultaneously, so their intersection has probability 0.

8 Given $P(A) = 0.6, P(B) = 0.3,$ and $P(A \cup B) = 0.7$, are events A and B independent?

a. **No, because** $P(A \cap B) = 0.20 \neq 0.18$ ✓

b. Yes, because $P(A \cap B) = 0.18$

c. Yes, because they are mutually exclusive

d. Cannot be determined

$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.6 + 0.3 - 0.7 = 0.20$. For independence, $P(A)P(B) = 0.6 \times 0.3 = 0.18$. Since $0.20 \neq 0.18$, they are not independent.

9 What is Bayes' formula for $P(A \mid B)$?

a. $\frac{P(B \mid A)P(A)}{P(B)}$ ✓

b. $\frac{P(A \mid B)P(B)}{P(A)}$

c. $\frac{P(A)P(B)}{P(A \cap B)}$

d. $P(A) + P(B)$

Bayes' theorem expresses conditional probability as $P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$.

Section 4: Combinatorial Probability & Geometric Probability

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10 A point is chosen at random inside a square of side 10. What is the probability that it falls inside the inscribed circle?

a. $\frac{\pi}{4}$ ✓

b. $\frac{\pi}{2}$

c. $\frac{\pi}{8}$

d. $\frac{1}{2}$

Area of square $= 10^2 = 100$. Inscribed circle radius $= 5$, Area $= \pi(5^2) = 25\pi$. Probability $= \frac{25\pi}{100} = \frac{\pi}{4} \approx 0.785$.

11 From a standard deck of 52 cards, what is the probability of being dealt a 5-card hand containing all 4 aces?

a. $\frac{48}{\binom{52}{5}}$ ✓

b. $\frac{4}{\binom{52}{5}}$

c. $\frac{1}{\binom{52}{5}}$

d. $\frac{48}{52}$

There is $\binom{4}{4} = 1$ way to select all 4 aces, and $\binom{48}{1} = 48$ ways to select the 5th card: $\frac{48}{\binom{52}{5}}$.

12 Two real numbers x and y are chosen independently from $[0, 1]$. What is the probability that $x + y \leq 1$?

a. $\frac{1}{2}$ ✓

b. $\frac{1}{4}$

c. $\frac{3}{4}$

d. $\frac{1}{3}$

The total sample space is a unit square of area 1. The region $x + y \leq 1$ is a right triangle with vertices $(0, 0), (1, 0),$ and $(0, 1)$, having area $\frac{1}{2}(1)(1) = \frac{1}{2}$.

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